

機械学習を用いた因果推論

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因果推論 at glance

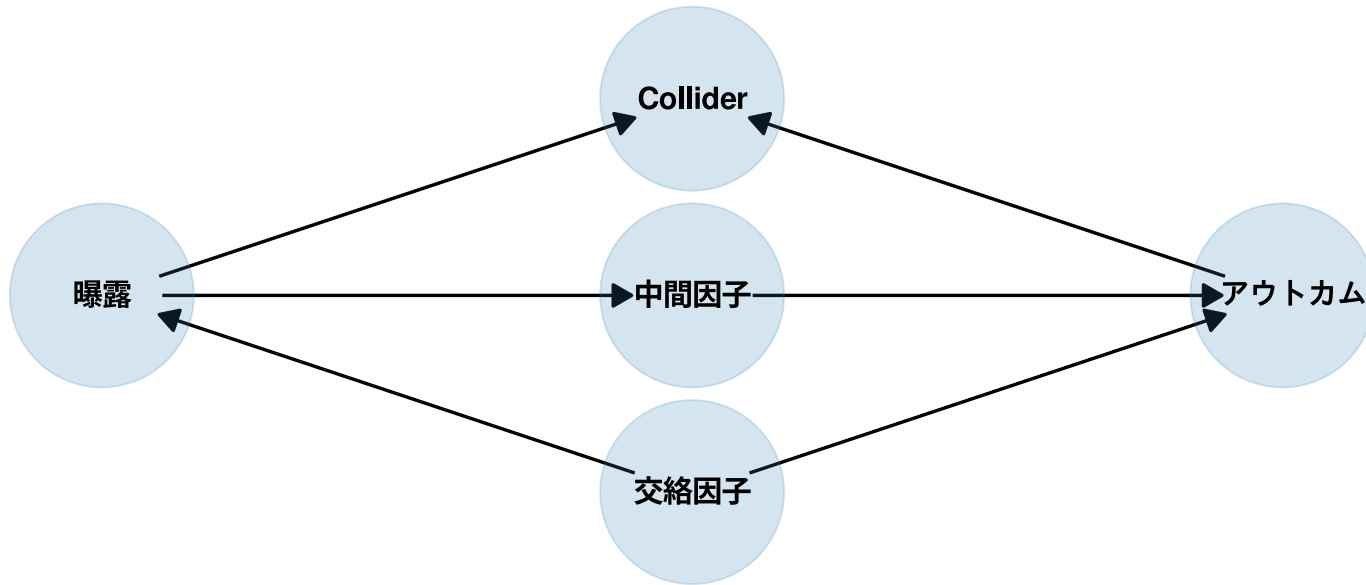
因果関係

- ▶ 医学研究の第一の目標と言っても過言ではない
- ▶ A の薬を使うと患者の予後は良くなるか？
- ▶ 実際は口で言うほど簡単ではない

因果関係の考え方

- ▶ 大きく分けて 2 つの考え方がある
 - 潜在的アウトカムを考える Rubin 流
 - DAG を作り、do operator を使う Pearl 流
- ▶ 詳細は省くが、今回は Pearl 流の DAG を中心に考える

DAG の基本



通常 of 統計学的手法

- ▶ 通常は交絡因子の調整により因果関係を推測する

交絡因子の調整の方法

- ▶ 回帰
- ▶ Matching
- ▶ 層別化

例えば単純な回帰だと

term	Coef	Std.Err	t	P
Intercept	-1174.13	2456.31	-0.48	0.63
treat	1548.24	781.28	1.98	0.05
age	12.98	32.49	0.4	0.69
educ	403.94	158.91	2.54	0.01
race_hispan	1739.54	1018.52	1.71	0.09
race_white	1240.64	768.76	1.61	0.11
married	406.62	695.47	0.58	0.56
nodegree	259.82	847.44	0.31	0.76
re74	0.3	0.06	5.09	0
re75	0.23	0.1	2.21	0.03

Matching だと

- ▶ 一番有名なのは Propensity score matching

```
treat age educ ... race_hispan race_white propensity_score
19    1  26  12 ...         0         0         0.619048
42    1  17   8 ...         0         0         0.619048
156   1  18   9 ...         0         0         0.217391
111   1  17   9 ...         1         0         0.217391
148   1  26  11 ...         0         0         0.217391
```

[5 rows x 11 columns]

matched size: (296, 11)

ATT (diff in means on matched sample): 1008.6141578378374

DAG つき

Model to find the causal effect of treatment ['treat'] on outcome ['re78']

Estimand type: EstimandType.NONPARAMETRIC_ATE

Estimand : 1

Estimand name: backdoor

Estimand expression:

$$\frac{d}{d[\text{treat}]} \left(E[\text{re78} | \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married}] \right)$$

Estimand assumption 1, Unconfoundedness: If $U \rightarrow \{\text{treat}\}$ and $U \rightarrow \text{re78}$ then $P(\text{re78} | \text{treat}, \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married}, U) = P(\text{re78} | \text{treat}, \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married})$

Estimand : 2

Estimand name: iv

Estimand expression:

DAG つき

[

|

d

/

d

\

]

-1

E|

(re₇₈)

|

[

d[_{race_{hispan}}

race_black]

\

d[_{race_{hispan}}

race_black]

]

]

Estimand assumption 1, As-if-random: If $U \rightarrow \rightarrow re_{78}$ then $\neg(U \rightarrow \rightarrow \{race_hispan, race_black\})$

Estimand assumption 2, Exclusion: If we remove $\{race_hispan, race_black\} \rightarrow \{treat\}$, then $\neg(\{race_hispan, race_black\} \rightarrow re_{78})$

Estimand : 3

Estimand name: frontdoor

No such variable(s) found!

Estimand : 4

Estimand name: general_adjustment

Estimand expression:

d

DAG つき

—————(E[re78|age,educ,nodegree,re75,married])

d[treat]

Estimand assumption 1, Unconfoundedness: If $U \rightarrow \{\text{treat}\}$ and $U \rightarrow \text{re78}$ then $P(\text{re78} | \text{treat}, \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married}, U) = P(\text{re78} | \text{treat}, \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married})$

*** Causal Estimate ***

Identified estimand

Estimand type: EstimandType.NONPARAMETRIC_ATE

Estimand : 1

Estimand name: backdoor

Estimand expression:

d

—————(E[re78|age,educ,nodegree,re75,married])

d[treat]

DAG つき

Estimand assumption 1, Unconfoundedness: If $U \rightarrow \{\text{treat}\}$ and $U \rightarrow \text{re78}$ then $P(\text{re78} | \text{treat}, \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married}, U) = P(\text{re78} | \text{treat}, \text{age}, \text{educ}, \text{nodegree}, \text{re75}, \text{married})$

Realized estimand

b: re78~treat+age+educ+nodegree+re75+married

Target units: ate

Estimate

Mean value: 973.336502850163

Refuting Estimates: 0%|██████████[32m ██████████[0m| 0/100 [00:00<?, ?it/s]

Refuting Estimates: 11%|██████████[32m#1 ██████████[0m| 11/100 [00:00<00:01, 71.56it/s]

Refuting Estimates: 29%|██████████[32m##9 ██████████[0m| 29/100 [00:00<00:00, 123.53it/s]

Refuting Estimates: 46%|██████████[32m###6 ██████████[0m| 46/100 [00:00<00:00, 141.33it/s]

Refuting Estimates: 64%|██████████[32m####4 ██████████[0m| 64/100 [00:00<00:00, 153.89it/s]

DAG つき

Refuting Estimates: 84%|██████████4 ██████████| 84/100 [00:00<00:00, 168.28it/s]

Refuting Estimates: 100%|███████████| 100/100 [00:00<00:00, 153.62it/s]

Refute: Add a random common cause

Estimated effect:973.336502850163

New effect:973.3365028501628

p value:1.0

Doubly robust estimation

DR ATE: 1394.87

SE: 926.77

95% CI: (-421.60, 3211.33)

ここらへんの問題

- ▶ Misspecification の問題
- ▶ Estimand の問題 = 治療効果の異質性

機械学習 at glance

代表的な機械学習

- ▶ Logistic 回帰
- ▶ tree-based model
 - Random forest
 - XGBoost
- ▶ Neural network

疑問

- ▶ 機械学習を用いて因果関係を推測出来るんじゃないか？

機械学習を用いた因果推論の Pros and Cons

▶ Pros

- モデルの Misspecification を避けられる
- 因果関係の異質性を柔軟に捉えられる

▶ Cons

- Overfitting の問題
- 解釈が難しい時がある
- 信頼区間が出しにくい

```
<econml.metalearners._metalearners.SLearner object at 0x70e5417e2d50>
```

```
[-1530.55967406  369.38136095 -1198.06865549  -86.11625054  
-833.38280696  171.70918773   0.          171.70918773  
 319.60055729  3232.14589762 -1326.64678003   62.06342827  
 973.47414377  503.67373985 -498.5500903  -128.93867654]
```

機械学習を用いた因果推論の Pros and Cons

```
485.25799913 489.9931725 1943.89516126 -269.05807531
2651.3162438 251.02443635 -3528.65968649 -591.89366066
436.53410022 32.09860508 -149.43314728 214.47455118
-86.11625054 32.09860508 3232.14589762 32.09860508
664.43858698 215.4353309 3451.76242614 -3558.48624607
-1148.88963224 2872.53054803 524.46483345 -544.94289508
632.3399819 -545.38598944 2651.3162438 144.60637046
-20.61071748 95.79529433 -86.11625054 -544.94289508
179.0257057 -2760.46335128 -957.9515668 1897.03733375
650.11851089 189.4222147 29.11044322 848.41449744
-352.45433824 2642.76190818 -1788.65634271 -988.00134014
-1198.06865549 3232.14589762 -522.42034513 -325.05097239
32.09860508 -167.70521158 2297.82108649 18.95269302
-1323.72233293 -35.38360418 74.66266901 -263.26905945
-289.37725534 3185.71468272 -122.99846869 -86.11625054
-128.93867654 1179.17631607 -2787.07045179 1789.97076026
32.09860508 -263.26905945 -843.00231034 -1539.13799559
```

機械学習を用いた因果推論の Pros and Cons

```
125.67766335 425.69190007 2199.59715977 -2259.89874238
2651.3162438 -1539.13799559 261.5062125 -495.49910709
-1218.45302373 -1539.13799559 2297.82108649 278.45737341
1861.49985719 664.43858698 95.79529433 2420.81146399
903.14992197 -329.34261136 894.79147023 194.23270811
-1179.12813405 171.70918773 171.70918773 -1089.00156337
341.10564214 369.38136095 -591.89366066 -86.11625054
-1131.29106563 -410.70666473 -20.61071748 -591.89366066
207.64243233 -86.11625054 319.60055729 2063.90546808
-2635.07325719 -662.79260714 715.06165279]
```

Scratch for S-learner

モデルの学習を開始します...

```
LGBMRegressor(random_state=42, verbosity=-1)
```

学習が完了しました。

テストデータのRMSE: 0.6124

平均処置効果の推定値: 749.0242

	treat	age	married	educ	...	effect_0	treat_1	effect_1	treat_e	>
0	1	37	1	11	...	2437.328621	1	6254.264565	3816.9	>
1	1	22	0	9	...	5855.361897	1	6215.004201	359.6	>
2	1	30	0	12	...	1654.736704	1	5863.334241	4208.5	>

Scratch for S-learner

```
3    1  27    0  11 ... 5778.341560    1 10567.286106  4788.9 >
4    1  33    0   8 ...  895.305362    1  3814.387914  2919.0 >
```

```
>
```

```
[5 rows x 15 columns] >
```

```
> ffct
```

```
> 35943
```

```
> 42304
```

```
> 97538
```

```
> 44546
```

```
> 82552
```

```
>
```

```
>
```

```
<econml.metalearners._metalearners.TLearner object at 0x70e540d86420>
```


Scratch for S-learner

T learnerでの局所治療効果

```
[ -6650.98045077 -1118.6693171 7910.85611871 -4644.60006793
-5193.61671006 -1265.01628653 -1076.66778986 -1884.26404861
1564.51358604 6387.09977938 -17618.45105122 -3612.68936737
573.13368963 16200.95698813 -3859.86899084 336.54393078
2849.52966914 5133.3472875 4362.02429344 -4925.08721332
-3299.94537519 -13131.62022045 -12833.90437168 -8136.22119316
2871.03924138 -1936.90571189 -10729.46961207 -550.75806049
-4162.5337728 -16485.71427559 4827.58362554 -9824.65040004
8204.3069157 1938.90093947 11337.47819452 -3406.60012456
-3963.31021415 2860.03677434 6471.99906902 -6261.968898
917.83931442 -1937.86327265 13587.22837397 1821.31726953
865.44040096 2312.87813147 934.3501449 -16426.57428451
594.8779234 3585.2194147 -4277.95596726 2136.27681753
7125.14806731 2138.29716778 -3883.91378688 -5218.15875866
```

Scratch for S-learner

```
-7799.68434228 4075.03162163 -5429.40421769 -9084.61075982
-8857.81565801 5029.10118391 -9352.85425194 -267.04791813
9704.22887422 -4425.27599461 2037.97086778 -9330.43734666
3460.91088841 5143.14296934 706.29492043 -8247.97419454
-621.94188682 9497.28415412 7091.82273943 -4162.5337728
-365.05015182 4028.98226432 4498.00494588 4616.20332428
-4302.66307485 -831.07548236 -3761.67302329 -4863.79074393
2338.24688147 8510.37044263 -4976.29005251 -3629.85675494
7007.06944427 -5855.50651842 4731.59503049 -3063.50546107
-912.18517109 -15568.23504452 2641.3341303 -5453.70340661
10867.85011588 5753.16004211 -1456.64247255 3368.49826808
-3180.22245828 2500.74116673 4251.33460448 -1130.11883784
-803.66883908 -3089.84565141 -2072.82922894 1787.99070559
-1960.36916504 4389.53911275 -9042.29350198 -4162.5337728
117.86381757 730.53927385 -3415.59825 -4492.28134015
-5048.95189277 -4162.5337728 3750.17902964 2291.90717807
3355.0384531 589.82480352 2833.65279389]
```

Scratch for S-learner

```
<econml.metalearners._metalearners.XLearner object at 0x70e541a4f020>
```

```
<econml.metalearners._metalearners.DomainAdaptationLearner object at 0x70e5417 >
```

```
> e3470>
```

```
<econml.dr._drlearner.DRLearner object at 0x70e5404bbe60>
```

機械学習の因果推論における使い方

- ▶ 交絡因子の調整で機械学習を用いる
 - AIPW, tlme など
- ▶ 因果をダイレクトに推論する学習機を作る
 - Meta-learner
- ▶ CATE を測定する
 - Causal forest など

因果 forest

- ▶ HTE を計測するのに有名
- ▶ Ashley et al 2019

Survival model

	id	study	rx	sex	age	obstruct	perfor	adhere	nodes	status	differ	extent
1	1	1	Lev+5FU	1	43	0	0	0	5	1	2	3
2	1	1	Lev+5FU	1	43	0	0	0	5	1	2	3
3	2	1	Lev+5FU	1	63	0	0	0	1	0	2	3
4	2	1	Lev+5FU	1	63	0	0	0	1	0	2	3
5	3	1	Obs	0	71	0	0	1	7	1	2	2
6	3	1	Obs	0	71	0	0	1	7	1	2	2

	surg	node4	time	etype
1	0	1	1521	2
2	0	1	968	1
3	0	0	3087	2
4	0	0	3087	1
5	0	1	963	2
6	0	1	542	1

Survival model

```
control treated
[1,] -566.2354 566.2354
[2,] 159.5300 -159.5300
[3,] -565.7077 565.7077
[4,] -553.9295 553.9295
[5,] -334.3212 334.3212
[6,] -1654.1933 1654.1933
```

Survival model

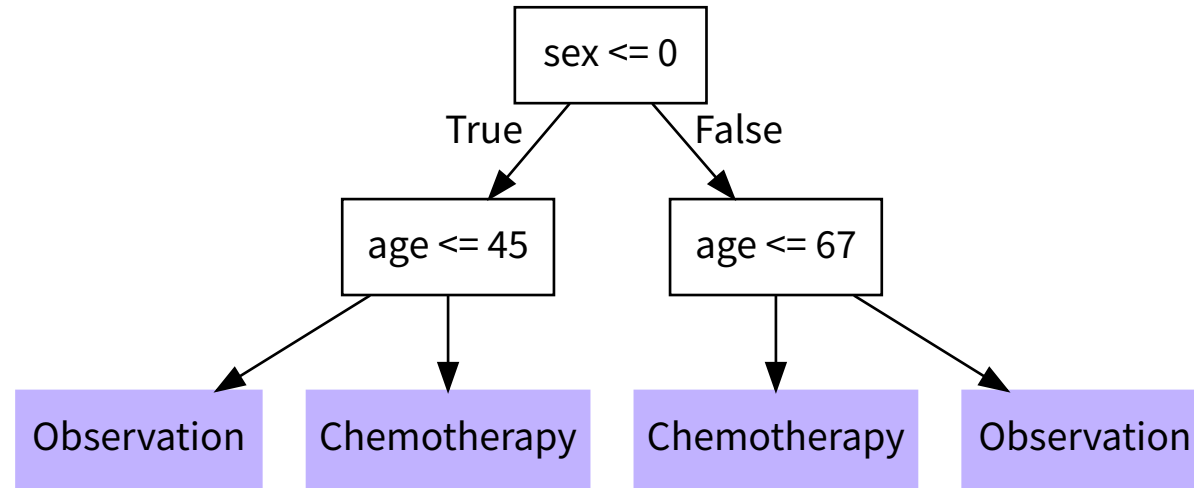


Figure 1: Treeplot